Spatial-filter refitting improves limited-label EEG decoding beyond classifier recalibration

Running head: Spatial refitting with limited calibration labels

Ziyuan Zhu*

College of Artificial Intelligence Medicine, Chongqing Medical University, Jinyun Campus, No. 61 Daxuecheng Middle Road, Shapingba District, Chongqing 401331, China

***Corresponding author:** Ziyuan Zhu; zzy2630816871@gmail.com

ORCID: https://orcid.org/0009-0005-1153-4926

Abstract

Objectives: To quantify the additional value of spatial-filter refitting beyond classifier recalibration when approximately 20 current-session labels are available.

Methods: We compared frozen common spatial patterns and linear discriminant analysis (CSP/LDA), classifier-only recalibration and full refitting in 18 participants across six Kumar2024 sessions. The predefined primary cohort contained 12 participants. Both updating methods used identical source and current-session labels. Later component experiments used the six previously exposed development participants.

Results: Full refitting achieved 65.36% balanced accuracy, exceeding freezing by 7.42 percentage points (95% participant-bootstrap interval 2.49–12.48). The secondary, matched-label full-minus-classifier-only gain was 5.47 points (2.17–8.91), accounting for 73.7% of the mean total gain. It was positive in ten of 12 participant averages and all five target-session averages. Neural adaptation with fixed normalization exceeded adaptive normalization by 1.50–1.91 points at the 20-label cap, with different smaller-budget profiles. The geometric candidate underperformed ordinary tangent refitting and Euclidean-aligned CSP; source-functional constraints added little benefit.

Conclusions: Spatial-filter refitting supplied most of the observed recalibration gain under matched labels. The component comparisons identify spatial updating and normalization mode as distinct design choices for limited-label longitudinal decoding.

Keywords: batch normalization; brain–computer interfaces; covariance alignment; motor imagery; transfer learning

Introduction

A longitudinal motor-imagery brain–computer interface (BCI) must use a short run of new labeled EEG effectively. Between-session changes can affect signal distributions, spatial representations and class decisions. Calibration can update the classifier, the representation, or both, with each fitted component depending on a small number of new observations [1].

Common spatial patterns (CSP) and linear discriminant analysis (LDA) make these choices directly testable [2, 3]. CSP extracts class-discriminative spatial power features; LDA maps them to decisions. Re-estimating LDA changes classification in the source feature space. Refitting both stages also changes that space. When the two updates

receive identical labels, their difference quantifies the added spatial update within the pipeline.

Previous research has established classifier adaptation, adaptive CSP, data-space transformations and distribution alignment [4–8]. The Kumar acquisition study combined covariance recentering with a Riemannian classifier, added supervised prototype updates, and examined a recentering-based CSP extension [9, 10]. These precedents motivate a component question: how much does relearning spatial filters contribute when classifier recalibration already receives the same new labels?

Neural adaptation introduces further choices. Updating weights can accompany changes in normalization statistics, affine parameters and Dropout. Adaptive batch normalization (BN) is established [11], and longitudinal EEG studies have combined fine-tuning with alignment and normalization updates [12]. Weight and functional constraints offer other ways to preserve source information [13–15]. Separating these actions helps explain which part of adaptation delivers a gain.

We first compared frozen CSP/LDA, LDA-only recalibration and full refitting across six Kumar2024 sessions. Both updates used 40 source-training labels and 19 or 20 current-session labels. The predefined primary contrast was full refitting versus freezing in a 12-participant cohort; full versus LDA-only was secondary. We then examined conventional alignment, geometric estimation and controlled neural modes in the six previously exposed development participants. The paper connects the matched-label spatial gain to its prediction transitions and longitudinal consistency, while the later experiments locate useful normalization choices and the limits of added candidate machinery.

Materials and methods

Data source and ethics

We analyzed the public Kumar2024 EEG derivative, NEMAR nm000177/v1.0.5 [16], associated with the original study and its correction [9, 10]. The original study reported University of Texas at Austin ethical approval under protocol 2020-03-0073 and written informed consent before data collection.

This secondary analysis was exempt from additional ethical review under Article 32, item (1), of China's *Measures for Ethical Review of Life Science and Medical Research Involving Human Participants* (2023) [17]. It used lawfully obtained publicly available data, caused no harm to participants, and did not involve sensitive personal information or commercial interests.

Dataset metadata were accessed on 10 October 2026, and the selected EEG recordings and accompanying metadata were retrieved on 11 October 2026. This secondary analysis used the released coded participant identifiers; no participants were contacted and no re-identification was attempted. The author had no access to information that could identify individual participants during or after data collection. Original collection approval and consent and the regulatory basis for this secondary analysis are documented separately in the retained provenance materials.

Windows, preprocessing and chronological information access

The derivative contains 18 participant identifiers, six sessions and 360 EDF recordings. Of 7,200 exported left/right events, 7,156 provided complete five-second windows; 44 reached a recording boundary and were excluded without padding. All 44 occurred at the last event of an online recording.

The original codebook assigns 769/770 to left/right cues and 7691/7701 to task starts [18]; the documented MOABB adapter maps cues to a [0, 5]-second interval [19]. The original procedure included a 1.5-second cue preceding feedback and variable online feedback duration. Our endpoint is therefore task-label decoding over released cue-anchored windows. Full trial-level phase composition remained unresolved.

Preprocessing used common-average referencing across 22 channels, fourth-order 8–30 Hz zero-phase Butterworth filtering, and resampling from 512 to 128 Hz, yielding 22×640 float32 values in volts. Learned transforms used training partitions.

Participants sub-1, 4, 7, 10, 13 and 16 formed the development group; the remaining 12 formed the original primary cohort. Session 0/runs 0–1 supplied 40 source-training windows, run 2 supplied 20 source-validation windows, and run 3 supplied source query. In later sessions, run 0 supplied support and subsequent whole runs query; updates used current-session labels without accumulating previous targets. The original query set contained 4,290 complete events, including 360 source and 3,930 target events; 30 partial-query exclusions remained fixed.

The existing 250 µV raw peak-to-peak flag marked 1,512/7,156 complete windows. The main analysis retained them. Sensitivity analysis required both classes in every target session.

Original component comparison

The conventional decoder used four-component CSP, Ledoit–Wolf regularization and automatic-shrinkage LDA [20]. Frozen decoding used 40 source-training trials. LDA-only retained source CSP and refitted LDA on source training plus current support; full refitting re-estimated both components on that same union. Their difference measures the added spatial update and classifier fit in the revised feature space.

The original EEGNet reference used three fixed seeds per participant, source normalization, 40 training labels and 20 source-validation labels for epoch selection [21, 22]. All 54 models and seed outcomes were retained. The HPC extension used the separately specified compact architecture below.

Prediction-transition analysis counted errors corrected by method B relative to A and the reverse changes, balancing the two classes. The difference between these contributions equals BA(B)−BA(A) exactly. This post-hoc analysis, class recalls, session profiles and participant-deletion means explain the original component endpoint.

Discovery extensions and frozen comparison roles

V2 and V3 used the same six previously exposed development participants, chronological whole-run partitions, scheduled seeds and query identities. Current support was capped at 4, 8, 12, 16 or 20 complete trials, retaining the observed class order. Within

each frozen execution, query was excluded from fitting and selection; the design of later rounds was informed by earlier results.

V2 tested source–target covariance mixing against EA-CSP. V3 fixed two cap-20 primary candidate contrasts before execution: uncertainty-aware prediction minus ordinary tangent refitting, and source-functional-constrained minus ordinary full neural adaptation within each source regime. BN, Dropout and dose comparisons were exploratory mechanisms. The full matrix included 10,350 target and 42 source prediction files, all nested within the same six participants; details are in the supplement.

Neural update-mode experiment

The compact HPC EEGNet used eight temporal filters, depth multiplier two, temporal kernel 64 and source Dropout 0.25. Individual sources reused 18 V2 checkpoints, each with 40 training and 20 validation labels. Three pooled sources, one per seed, used 240 training and 120 validation examples from the same six participants, changing source-data amount and cross-participant composition.

Source training used 100 epochs with validation cross entropy selecting the model. Optimizer adaptations used current support, 20 epochs, batch 16, AdamW, learning rate 10^{-4}, weight decay 10^{-4}, a cosine schedule and no automatic mixed precision. A separate sampler matched support order. Query inference used evaluation mode.

Eight policies covered freezing, head-only, full fixed-BN, full adaptive-BN, adaptive-BN plus Dropout, BN-affine-only, BN-statistics-only and functional-constrained full updates. Dropout was off except in its explicit arm. Fixed BN used source running statistics during adaptation and inference; adaptive BN used batch statistics during adaptation and updated the inference state. This contrast changes normalization during optimization and the resulting buffers, with potentially different weight trajectories.

The functional policy bounded mean teacher-to-candidate KL on 40 source-training anchors at 0.01, with additional doses 0.001 and 0.1. Bounded backtracking restored parameters and optimizer state after a rejected proposal [14, 15].

Geometric estimator and matched controls

Trace-normalized trial covariances used fixed 0.1 shrinkage and a source-training-only log-SPD reference, giving 253 features for 22 channels. Class means were represented by common location $m=(\mu_0+\mu_1)/2$ and half contrast $h=(\mu_1-\mu_0)/2$. The candidate estimated separate noise-corrected shift scales, matrix mean-noise covariances and a linear-Gaussian posterior. Prediction added class-mean posterior covariance to fixed source within-class covariance.

Controls covered frozen source prediction, ordinary pooled means, tangent refitting, candidate point prediction, isotropic noise, coupled shift scales, and removal of source noise or shift variance. Missing support classes followed the fixed masked-observation rule. Equations and assumptions are in the supplement.

Saved synthetic experiments tested all nine methods across 384 known-mechanism conditions and 50 paired repetitions, varying dimension, sample sizes, shift, anisotropy and serial correlation. These results were analyzed separately from EEG.

Participant-level analysis and saved-result verification

BA averaged the two true-class recalls. Seeds were averaged within session, five target sessions within participant, then participants within cohort. Paired contrasts used matched endpoints and 10,000 participant-bootstrap resamples: seed 42 for the original comparison and 20261023 for the extensions. The six-person discovery intervals are pointwise summaries.

All scheduled seeds, sessions, caps and doses were retained; any missing required cell left the endpoint incomplete. Pooled-source intervals are conditional on the fitted shared teachers. Synthetic summaries equally weighted the fixed scenarios and paired draws. Exploratory EEG cross entropy clipped probabilities at 10^{-15}; simulation scoring used the same numerical floor.

Saved trial identities, labels and metrics were checked before aggregation. The revision reused predictions and histories; no model fitting or new inference was required. The supplement identifies the full result packet, code and provenance checks.

Results

Spatial refitting supplied the main matched-label gain

Full CSP/LDA refitting achieved 65.36% target BA in the original 12-participant primary cohort, compared with 57.94% for frozen decoding and 59.89% for LDA-only recalibration. The predefined primary full-minus-frozen gain was 7.42 percentage points (95% interval 2.49–12.48). With identical fitting labels, the secondary full-minus-LDA-only gain was 5.47 points (2.17–8.91), while LDA-only minus frozen was 1.95 points (−0.98–5.09). The spatial-component contrast accounted for 73.7% of the mean total gain (Table 1; Figure 1).

Table 1. Original target-query component comparison.

Cohort	Participants	Frozen CSP/LDA BA	LDA-only BA	Full CSP/LDA BA	Full − frozen	Full − LDA-only
Development	6	59.97%	64.37%	63.22%	+3.25 pp	−1.15 pp
Predefined primary cohort	12	57.94%	59.89%	65.36%	+7.42 [2.49, 12.48] pp	+5.47 [2.17, 8.91] pp
All participants, secondary	18	58.62%	61.38%	64.65%	+6.03 pp	+3.26 pp

The two updating methods used the same source-training/current-support union. Values average five sessions within participants. Full refitting versus freezing in the 12-participant cohort was the predefined primary comparison; full versus LDA-only was secondary.

The component gain reflected 16.04 BA points of corrected errors against 10.57 points of newly introduced errors on the same query trials. Mean left- and right-class recall increased by 6.16 and 4.78 points, respectively. These recall intervals individually included zero; their class-balanced mean gives the observed component endpoint. The added spatial update was positive in ten of 12 participant averages and every target-session cohort average: +3.52, +5.20, +4.13, +4.18 and +10.33 points. Every single-participant deletion mean remained positive, between +4.39 and +6.28 points.

Full refitting exceeded freezing for nine of 12 primary participants. The three decreases were −5.82, −0.67 and −0.20 points. In the ten participants retaining both query classes across all five unflagged sessions, full-minus-frozen was +5.96 points (0.99–11.24) and full-minus-LDA-only +3.82 (1.11–6.72). This query restriction left original training and support data unchanged. The retained cohort and amplitude-conditional endpoint differ from the complete primary analysis.

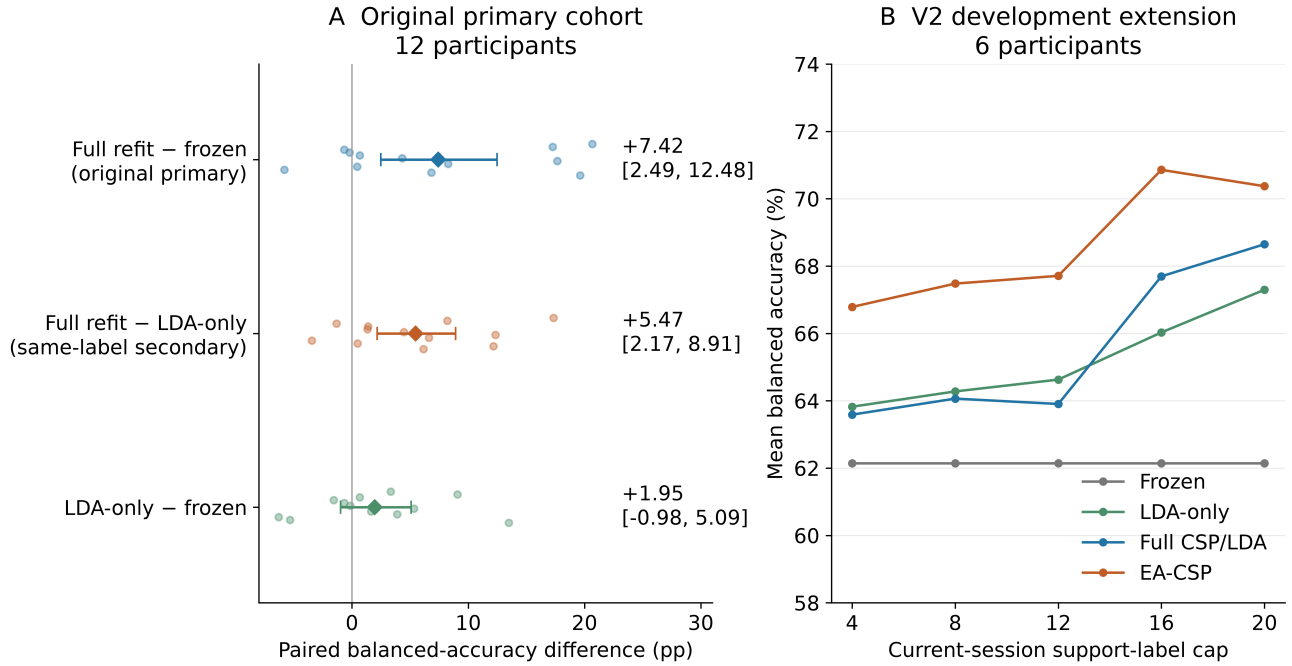


Figure 1. Spatial-component evidence and strong-reference context. Original primary-cohort component and total contrasts are displayed separately from the six-participant V2 budget profiles. Participant values, cohort means and participant-bootstrap intervals retain their original denominators and endpoint hierarchy. The original full-minus-LDA-only result compares matched labels; the V2 profiles compare separately versioned methods within the discovery cohort.

Strong alignment controls changed the interpretation of a positive frozen-reference gain

In the six-participant V2 extension at the 20-label cap, frozen classical decoding achieved 62.15% BA, LDA-only 67.30%, full CSP/LDA 68.65% and EA-CSP 70.37%. The covariance-mixture candidate achieved 69.24%. Its gain over frozen decoding was +7.09 points (4.12–11.55), but its prespecified comparison with EA-CSP was −1.14 points (−3.52–1.14). EA-CSP therefore provided the stronger conventional reference for the next ex-

periment.

This pattern motivated the more controlled V3 decomposition. V2 compared separately versioned conventional methods within the exposed development group.

The uncertainty-aware geometric candidate failed its strong comparisons

At the cap of 20 labels, the geometric candidate achieved 64.68% BA, versus 67.01% for ordinary tangent refitting and 70.37% for saved EA-CSP. Its prespecified primary contrast was −2.33 points (−4.41 to −0.24), and its strong-reference contrast was −5.69 points (−9.30 to −2.22; Table 2). Five of six participant differences were negative in both comparisons. All five support-cap mean contrasts versus tangent refitting were negative (Figure 2).

Table 2. V3 geometric comparison at the 20-label cap.

Method	Mean BA	Candidate − method, pp
Frozen source tangent model	60.36%	+4.32 [0.07, 8.61]
Pooled means with fixed source within-class covariance	64.08%	+0.60 [−3.09, 3.01]
Ordinary tangent refitting	67.01%	−2.33 [−4.41, −0.24]
Candidate means with point prediction	66.52%	−1.84 [−3.26, −0.46]
Full uncertainty-aware candidate	64.68%	Reference
Saved EA-CSP, strong reference	70.37%	−5.69 [−9.30, −2.22]

Ordinary tangent refitting was the frozen primary comparator. The same-query EA-CSP reference was imported from saved V2 predictions after ordered trial/label verification. All results use the same six discovery participants. Candidate ablations and every budget are retained in the supplementary methods and the machine-readable tables.

Some internal components carried useful information: separating common and discriminative variation and retaining source-mean noise improved the cap-20 candidate mean over their respective ablations. However, propagating the estimated posterior covariance into prediction reduced BA relative to using the same posterior means with point prediction. Estimated common-location variation exceeded discriminative-contrast variation in every cap-20 session, a descriptive feature of these support-derived estimates. The ablations separate the contributions of mean estimation, variation scales and predictive covariance.

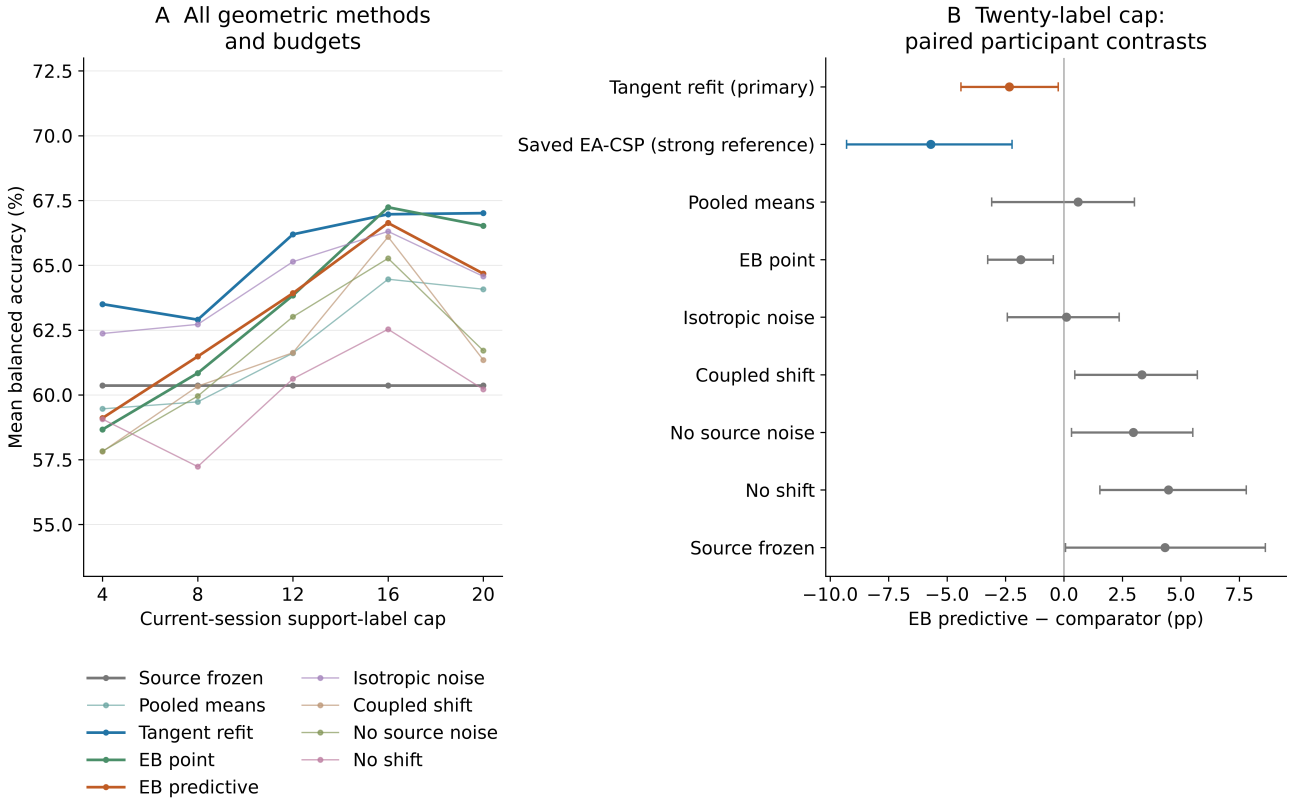


Figure 2. Geometric candidate, matched controls and ablations. Full support-budget profiles and cap-20 paired contrasts retain ordinary tangent refitting and EA-CSP as strong references. Ablations separate posterior mean estimation, prediction covariance, source-mean noise and common/contrast variation scales. All candidate-versus-ablation and strong-reference contrasts are displayed together; intervals are pointwise participant-bootstrap summaries from the exposed discovery group.

Fixed source normalization improved the 20-label neural comparison

At the 20-label cap, ordinary all-parameter adaptation with fixed BN achieved 57.51% BA for individual sources, versus 56.01% with adaptive BN, yielding +1.50 points (0.33–2.74). For pooled sources, the corresponding values were 56.22% and 54.31%, a difference of +1.91 points (0.76–3.24; Table 3). Five of six participant differences favored fixed BN in each regime; the remaining difference was negative for individual sources and approximately zero for pooled sources. All participant-deletion mean contrasts stayed positive at this cap.

Table 3. V3 neural component contrasts at the 20-label cap.

Contrast	Individual source models	Pooled source models	Role
Fixed-BN full update − adaptive-BN full update	+1.50 [0.33, 2.74] pp	+1.91 [0.76, 3.24] pp	Exploratory BN-mode comparison

Contrast	Individual source models	Pooled source models	Role
Fixed-BN full update − frozen	+0.77 [−0.16, 1.73] pp	+0.96 [−0.41, 2.77] pp	Update baseline
BN-statistics-only − frozen	−1.15 [−2.15, −0.12] pp	−0.30 [−0.68, 0.05] pp	Statistics-only comparison
Functional constraint − fixed-BN full update	−0.006 [−0.111, 0.092] pp	+0.029 [0.000, 0.088] pp	Prespecified neural primary contrast
Adaptive BN plus Dropout − adaptive BN	−0.15 [−0.40, 0.08] pp	+0.28 [0.04, 0.55] pp	Exploratory Dropout comparison

All intervals summarize six participants after averaging fixed seeds and target sessions. Both source regimes use the same participants. The functional-constraint row retains its frozen primary role and shows effects below 0.03 percentage points. The BN-mode contrast changes optimization-time normalization as well as the running state used at inference.

The cap-specific positive contrast accompanied fewer single-class query predictions. Among 90 nested seed-by-session outcomes, individual-source fixed-BN full updates had eight collapsed predictions versus 16 with adaptive BN; pooled-source counts were zero versus 18. These 90 seed/session observations are nested within six participants.

The budget curves show the scope of the finding (Figure 3). Individual-source fixed-minus-adaptive BN differences were −0.18, −0.34, −0.46, −0.04 and +1.50 points across caps 4–20. Pooled-source differences were +0.75, +0.01, +0.48, +1.31 and +1.91 points. The endpoint contrast emerged at the final support cap. With batch size 16, the final cap also changed the number and size of minibatches: up to 16 support trials supplied one batch per epoch, whereas 19/20 supplied two. Three BN modules recorded 60 versus 120 cumulative batch-counter increments over 20 epochs. Cross-budget changes combine sample quantity, class composition and the normalization-update schedule.

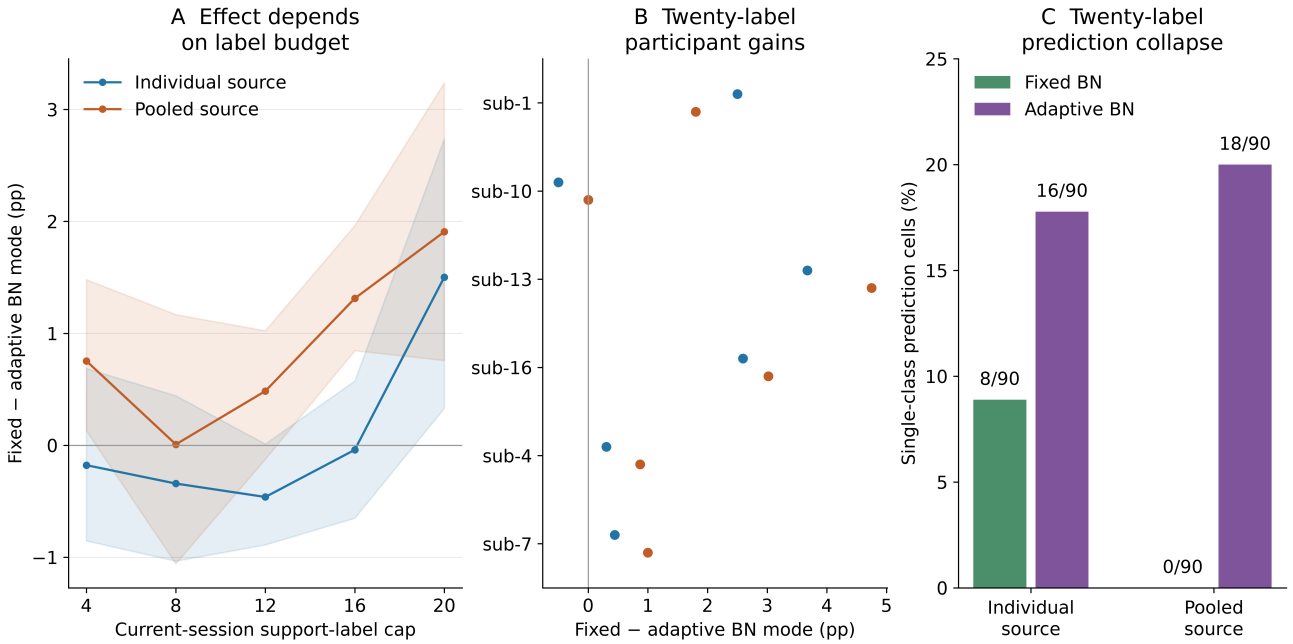


Figure 3. Neural normalization mode across support budgets. Fixed-minus-adaptive BN contrasts are shown for both source regimes across all five caps, together with the cap-20 participant contrasts and single-class prediction counts. Fixed and adaptive BN use the same current support, initialization, optimizer budget and Dropout-off setting within each comparison. They differ in normalization during optimization and the resulting inference state. Collapse counts are nested seed/session observations; participant-level summaries define the uncertainty intervals.

Source pooling and functional preservation did not deliver additional target gains

Frozen individual-source models achieved 56.74% target BA, compared with 55.26% for pooled-source models, a pooled-minus-individual difference of −1.48 points (−2.90 to −0.20). Source-query BA was 56.11% and 57.78%, respectively. Weak source discrimination was already present before target-session transfer. The pooled-source condition yielded a lower target mean in this architecture and training regime.

The three pooled models selected epochs 14, 20 and 17 by validation loss. Their final training losses were approximately 0.20–0.22, while final validation losses were 0.91–0.96, above the selected validation losses near 0.68. These histories show training-loss reduction alongside weak held-out generalization within the source domain.

At the core KL bound 0.01, the functional constraint changed the cap-20 mean relative to ordinary fixed-BN adaptation by −0.006 points for individual sources and +0.029 points for pooled sources. At bound 0.1, BA matched the ordinary update in both regimes; bound 0.001 produced differences of −0.22 and −0.19 points, respectively, with intervals crossing zero. The complete dose profiles and accepted/rejected-step histories are retained.

Mechanism simulations tested the estimator beyond the EEG comparison

The full synthetic grid contained 172,800 numeric method-by-condition-by-repetition results. Equal-grid BA was 70.36% for the candidate and 76.19% for tangent refitting, a difference of−5.83 points; 376 of 384 scenario means favored refitting. Candidate feature-mean squared error was also larger. Adding posterior covariance to identical candidate means reduced mean cross entropy by 0.341 but changed BA by−0.04 points. Thus, the probability-score improvement and classification endpoint moved differently (Figure 4).

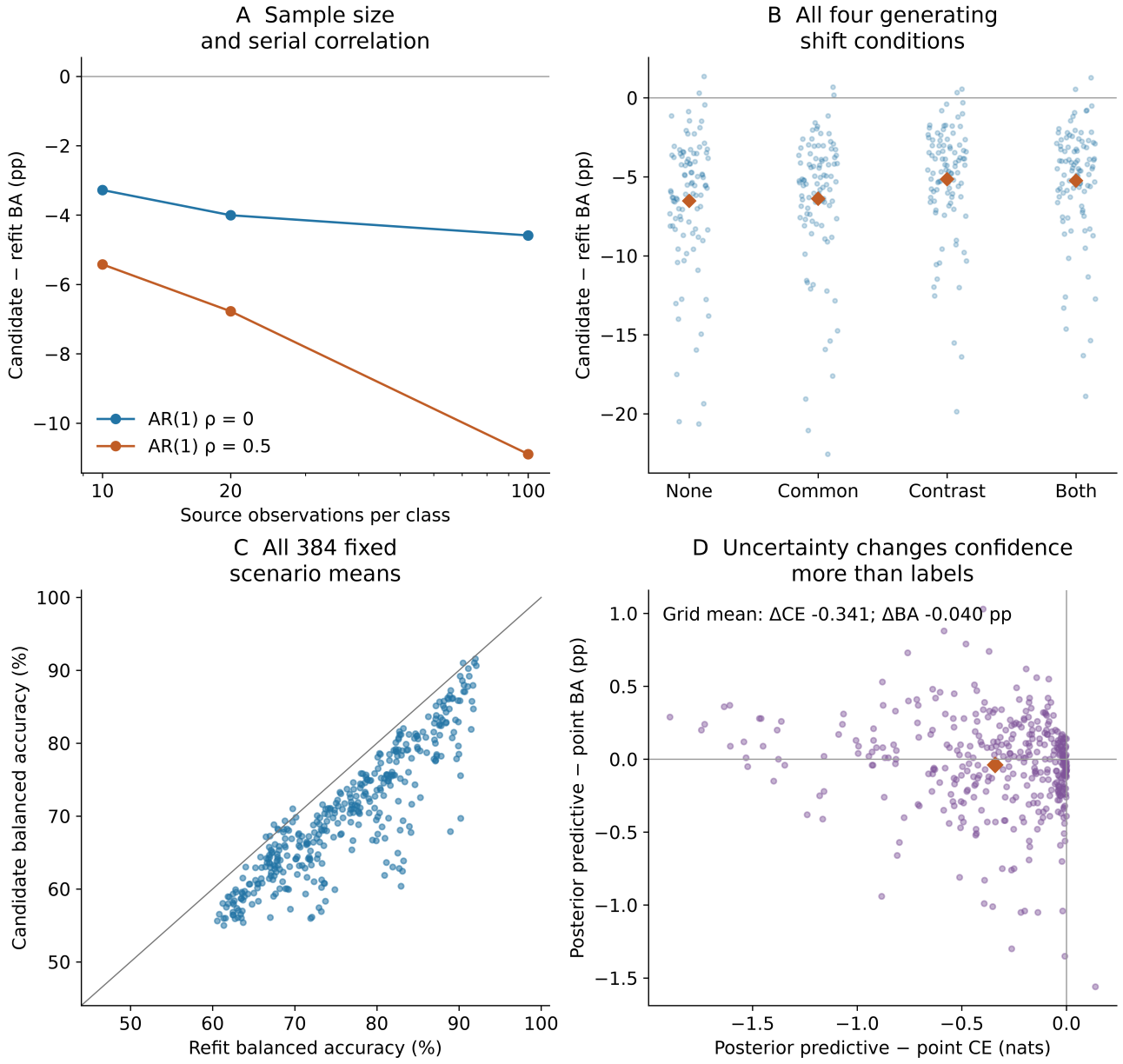


Figure 4. Mechanism simulations and predictive uncertainty. All 384 conditions retain 50 paired repeats. Panels show candidate-minus-refit BA by source size/correlation, all generating shifts, paired scenario BA, and posterior-versus-point changes in cross entropy and BA. Synthetic results are separate from human EEG endpoints.

Discussion

Updating the right component was more useful than updating every component

The matched-label comparison identifies a substantial spatial-component gain. Full CSP/LDA refitting added 5.47 BA points beyond classifier-only recalibration in the original primary cohort and supplied 73.7% of the mean total improvement over freezing. Holding the fitting examples constant gives this result a clearer interpretation than an updated-versus-frozen comparison alone: the available labels supported a useful change in the spatial representation as well as in the class decision rule.

The neural extension adds a complementary observation. At the final support cap, fixed-source BN during full weight adaptation outperformed adaptive BN in both initialization regimes, with fewer collapsed seed/session predictions. This contrasts with the successful CSP spatial update. Together, the comparisons show why "adapt the representation" is too broad a prescription: a supervised spatial estimator and repeated neural normalization updates are different operations with different data requirements. The controlled comparison identifies which update supplied the gain in each evaluated setting.

The principal conventional result also has a direct trial-level explanation. The added spatial update corrected more class-balanced errors than it introduced, and its mean benefit involved both labels. Positive session means and participant-deletion summaries show that its cohort average was not supplied by a single participant or only one point in the session sequence. The participant and development-cohort results document variation around this positive primary-cohort mean.

Normalization mode is a concrete, budget-dependent design variable

The BN comparison holds source initialization, support, trainable parameters, Dropout and optimizer schedule fixed within each budget. It identifies an implementation choice often bundled into fine-tuning. Both modes use identical support and optimization budgets. Their cap-20 difference isolates the BN-mode choice within this training procedure.

Adaptive BN changes batch normalization during optimization and accumulates the running state used for evaluation; both can affect the resulting model. The small final minibatch, number of BN updates and class composition differ across the chronological prefixes. The full curves show a different sign for the individual-source comparison at smaller caps.

These details distinguish our information regime from reports using abundant target-domain statistics or continuous online adaptation [11, 12]. They also give a focused next hypothesis: when adaptation repeatedly revisits a short support run, normalization mode and exposure count should be controlled independently of label count. A follow-up can vary the BN update schedule while preserving the trained sources, support identities and comparison hierarchy.

Strong controls separated practical gains from candidate-method novelty

EA-CSP outperformed the V2 covariance-mixture candidate and the V3 uncertainty-aware candidate. The latter also underperformed ordinary tangent refitting in the same feature family. These comparisons distinguish the value of recalibration from the value of a particular new estimator.

The candidate separated shared class movement from discriminative change and propagated estimated mean uncertainty. Ablations showed contributions from its shift scales and noise estimates, while point prediction identified a cost of adding posterior covariance. Its conditional expected squared-error result concerns class-mean estimation under the Gaussian assumptions; classification remains an empirical endpoint.

Similarly, source-anchor KL was enforced but changed cap-20 BA by less than 0.03 points. Alignment, source regularization and functional preservation are established [7, 13–15]. The present evidence identifies their value through explicit information budgets and matched component comparisons.

Source quality and task measurement constrain interpretation

Original primary-cohort EEGNet BA was 52.92% on source query and 51.58% on target sessions. The extension's sources likewise showed weak held-out discrimination despite falling training losses. Source generalization therefore contributes to interpreting target performance alongside session transfer.

The original source/support probes also distinguished calibration agreement from correct task orientation. Of 30 development support sessions, 19 had positive source-score contrasts in both chronological halves, two were negative in both, and nine changed sign. Agreement can coexist with reversed orientation. These findings motivate testing support-driven selection among fixed component choices in an independent evaluation.

Scope and next validation

The study uses one public longitudinal cohort. The original primary hierarchy is retained; V2/V3, prediction transitions and sensitivity analyses used previously exposed data. Their intervals describe discovery-stage component effects. Pooled teachers share participants and change source-label budgets; the independent sample remains the participant cohort.

Cue-anchored windows may contain cue, imagery, feedback or post-task activity. Phase-specific decoding requires original marker-level reconstruction. Amplitude-restricted queries retained positive spatial gains in eligible participants, while training/support artifacts and other query artifacts remain possible. Zero-phase filtering makes the present endpoint offline.

Independent longitudinal validation should freeze the matched-label spatial comparison and normalization-mode contrast before accessing new query outcomes. The current findings identify a substantial spatial-refitting increment and a distinct, budget-dependent normalization effect.

Conclusions

Under matched source and current-session labels, spatial-filter refitting supplied most of the observed conventional-decoder recalibration gain. The added spatial update improved BA by 5.47 points beyond LDA-only recalibration in the primary cohort and accounted for 73.7% of the 7.42-point total mean gain. In the later six-person discovery experiment, fixed-source BN improved the final-cap neural comparison by 1.50–1.91 points,

with budget-dependent profiles. The uncertainty-aware and functional-constraint candidates added no strong-comparator benefit. The matched-label spatial gain identifies filter updating as the principal contributor to the original recalibration improvement, with normalization mode providing a distinct neural design variable.

Data and code availability

Source EEG is available as NEMAR nm000177/v1.0.5, DOI 10.82901/nemar.nm000177.v1.0.5, under CC BY 4.0 [16]. Code, configurations, original results and saved extension predictions are available at https://github.com/jackzhu119/few-shot-mi-eeg, research branch `research/paper2-hpc-v2-20261023`. The supplement identifies the fixed V3 packet and reproduction inputs. Original software uses MIT; manuscript, figures and derived results use CC BY 4.0 within the scope defined in LICENSING.md. Source-data and third-party terms are retained.

Acknowledgments

The author thanks the researchers who released Kumar2024. OpenAI ChatGPT/Codex assisted with code, saved-result checks, plotting code, reference checks, drafting, editing and document preparation. Checks included trial matching, checksum verification and endpoint reconstruction. The author is responsible for the content.

References

1. Lotte F, Bougrain L, Cichocki A, Clerc M, Congedo M, Rakotomamonjy A, et al. A review of classification algorithms for EEG-based brain–computer interfaces: a 10 year update. J Neural Eng 2018;15:031005. doi:10.1088/1741-2552/aab2f2

2. Ramoser H, Muller-Gerking J, Pfurtscheller G. Optimal spatial filtering of single trial EEG during imagined hand movement. IEEE Trans Rehabil Eng 2000;8:441–446. doi:10.1109/86.895946

3. Blankertz B, Tomioka R, Lemm S, Kawanabe M, Muller K. Optimizing Spatial filters for Robust EEG Single-Trial Analysis. IEEE Signal Process Mag 2008;25:41–56. doi:10.1109/msp.2008.4408441

4. Vidaurre C, Kawanabe M, Bünau P von, Blankertz B, Müller KR. Toward Unsupervised Adaptation of LDA for Brain–Computer Interfaces. IEEE Trans Biomed Eng 2011;58:587–597. doi:10.1109/TBME.2010.2093133

5. Arvaneh M, Guan C, Ang KK, Quek C. EEG Data Space Adaptation to Reduce Intersession Nonstationarity in Brain-Computer Interface. Neural Comput 2013;25:2146–2171. doi:10.1162/NECO_a_00474

6. Song X, Yoon S-C. Improving brain-computer interface classification using adaptive common spatial patterns. Comput Biol Med 2015;61:150–160. doi:10.1016/j.compbiomed.2015.03.023

7. He H, Wu D. Transfer Learning for Brain–Computer Interfaces: A Euclidean Space Data Alignment Approach. IEEE Trans Biomed Eng 2020;67:399–410. doi:10.1109/TBME.2019.2913914

8. Zanini P, Congedo M, Jutten C, Said S, Berthoumieu Y. Transfer Learning: A Riemannian Geometry Framework With Applications to Brain–Computer Interfaces. IEEE Trans Biomed Eng 2018;65:1107–1116. doi:10.1109/TBME.2017.2742541

9. Kumar S, Alawieh H, Racz FS, Fakhreddine R, Millán J del R. Transfer learning promotes acquisition of individual BCI skills. PNAS Nexus 2024;3:pgae076. doi:10.1093/pnasnexus/pgae076

10. Correction to: Transfer learning promotes acquisition of individual BCI skills. PNAS Nexus 2024;3:pgae137. doi:10.1093/pnasnexus/pgae137

11. Li Y, Wang N, Shi J, Liu J, Hou X. Revisiting Batch Normalization For Practical Domain Adaptation. 2016. Available: https://arxiv.org/abs/1603.04779

12. Wimpff M, Aristimunha B, Chevallier S, Yang B. Fine-Tuning Strategies for Continual Online EEG Motor Imagery Decoding: Insights from a Large-Scale Longitudinal Study. 2025 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) IEEE; 2025. pp. 1–7. doi:10.1109/EMBC58623.2025.11253543

13. Li X, Grandvalet Y, Davoine F. Explicit Inductive Bias for Transfer Learning with Convolutional Networks. Proceedings of the 35th international conference on machine learning PMLR; 2018. pp. 2825–2834. Available: https://proceedings.mlr.press/v80/li18a.html

14. Li Z, Hoiem D. Learning without Forgetting. 2016. Available: https://arxiv.org/abs/1606.09282

15. Schulman J, Levine S, Abbeel P, Jordan M, Moritz P. Trust Region Policy Optimization. Proceedings of the 32nd international conference on machine learning PMLR; 2015. pp. 1889–1897. Available: https://proceedings.mlr.press/v37/schulman15.html

16. Kumar S, Alawieh H, Racz FS, Fakhreddine R, Millan J del R. Multi-session longitudinal motor imagery EEG dataset (Kumar et al. 2024). NEMAR (Neuroelectromagnetic Data Archive; Tools Resource); 2026. doi:10.82901/nemar.nm000177.v1.0.5

17. National Health Commission of the People's Republic of China, Ministry of Education of the People's Republic of China, Ministry of Science and Technology of the People's Republic of China, National Administration of Traditional Chinese Medicine. Measures for the Ethical Review of Life Science and Medical Research Involving Humans. National Health Commission of the People's Republic of China; 2023. Available: https://www.gov.cn/zhengce/zhengceku/2023-02/28/content_5743658.htm

18. EEG event codebook: Dataset_Description.pdf. TransLearning repository [Internet]; GitHub; [cited 19 Oct 2026]. Available: https://github.com/neurosatya/TransLearning/blob/548470542e63f97ab777684c1f5e4f1cace4b8ed/Dataset_Description.pdf

19. Jayaram V, Barachant A. MOABB: trustworthy algorithm benchmarking for BCIs. J Neural Eng 2018;15:066011. doi:10.1088/1741-2552/aadea0

20. Ledoit O, Wolf M. A well-conditioned estimator for large-dimensional covariance matrices. J Multivar Anal 2004;88:365–411. doi:10.1016/s0047-259x(03)00096-4

21. Lawhern VJ, Solon AJ, Waytowich NR, Gordon SM, Hung CP, Lance BJ. EEGNet: a compact convolutional neural network for EEG-based brain–computer interfaces. J Neural Eng 2018;15:056013. doi:10.1088/1741-2552/aace8c

22. Schirrmeister RT, Springenberg JT, Fiederer LDJ, Glasstetter M, Eggensperger K, Tangermann M, et al. Deep learning with convolutional neural networks for EEG decoding and visualization. Hum Brain Mapp 2017;38:5391–5420. doi:10.1002/hbm.23730

Supporting information captions

Supplementary methods and data. Complete component definitions, mathematical assumptions, all methods/budgets/seeds, participant and scenario tables, original sensitivity analyses, prediction identities, provenance and saved-output reproduction instructions.